

ASTEROIDS

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ASTEROIDS ARE REMNANTS OF a failed attempt to form a rocky planet that would have been about four times as massive as Earth. They are dry, dusty objects and far too small to have atmospheres.

Over 200,000 have been discovered, although over a billion are predicted to exist. The astronomers who discover asteroids have the right to name them.

ORBITS

Most asteroids are found in a concentration known as the Main Belt, which lies between Mars and Jupiter, about 2.8 times farther from the Sun than Earth. Typically, they take between 4 and 5 years to orbit the Sun. The orbits are slightly elliptical and of low inclination. Even though the asteroids are all orbiting in the same direction, collisions at velocities of a few miles per second often take place. So as time passes, asteroids tend to break up. Some asteroids have been captured into rather strange orbits. The Trojans have the same orbital period as Jupiter and tend to be either 60° in front or 60° behind that planet. Then there are the Amor and Apollo asteroid groups (named after individual asteroids), with paths that cross the orbits of Mars and Earth, respectively. Aten asteroids have such

small orbits that they spend most of their time inside Earth's orbit. These three groups are classed as near-Earth asteroids. They can be dangerous, having the potential to hit Earth and cause a great deal of damage. Fortunately, this happens very rarely.

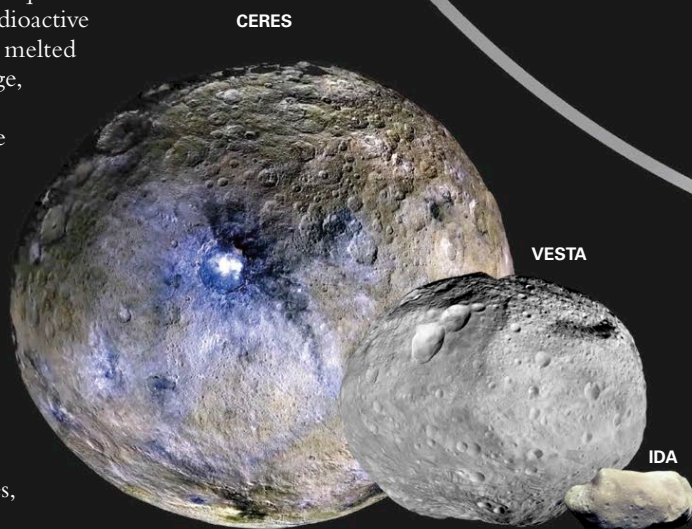
ASTEROID PATH

To picture stars, the Hubble Space Telescope scans the sky, keeping the stars stationary in the image frame. Asteroids, being much closer than the stars and in orbit around the Sun, form streaked trails (the blue line) during the exposure time.



STRUCTURES

At the dawn of the solar system, there existed quite a few asteroids nearly as large as Mars. The radioactive decay of elements within the asteroidal rock melted these large bodies and, during their fluid stage, gravity pulled them into a spherical shape before they cooled. Many of these have since been broken up or reshaped by collisions with other asteroids. Smaller asteroids, which cooled more efficiently than larger ones, did not reach melting point and retained a uniform rocky-metallic composition and their original irregular shape. There are three main compositional classes of asteroids. The vast majority are either carbonaceous (C-type) or silicaceous (S-type). The next most populated class is metallic (M-type). These classes correspond to carbonaceous chondrite (stony) meteorites, stony-iron meteorites, and iron meteorites.



CERES

VESTA

IDA

ASTEROID SHAPES AND SIZES

The largest asteroids, such as Ceres and Vesta, are nearly spherical, whereas smaller asteroids, such as Ida, are irregularly shaped. All asteroids have craters on their surfaces, but some areas have been sandblasted and smoothed by a multitude of minor collisions.

